# Climate Impacts

## Estimating the economic effects of climate

*Identification, measurement, and two applications*

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# Outline

## Dell, Jones & Olken (2014), *Journal of Economic Literature*

How the effect of climate on an economic outcome can be identified at all, and what has to be decided before any regression can be run.

## Dell, Jones & Olken (2012), *AEJ: Macroeconomics*

An application of that design, organised around a single testable question: does temperature affect the level of output or its growth rate?

## Jones, Moscona, Olken & von Dessauer (2026)

A recent paper arguing that the most common specification for nonlinear temperature effects is systematically biased, and proposing a correction.

# Part I

What do we learn from the weather?

# Climate and weather

$$y = f(C, X)$$

- $y$ is an economic outcome: income, mortality, agricultural yields, conflict.
- $C$ denotes the **climate**, understood as the full probability distribution from which daily weather is drawn.
- $X$ collects everything else that determines the outcome: institutions, technology, the capital stock, policy.

## Climate is a distribution; weather is a realisation of it

Rio de Janeiro's climate is what determines how you dress in general; the rain on a particular Tuesday is weather. The distinction matters because a regression of $y$ on observed weather estimates the effect of a realisation, whereas climate change is a change in the underlying distribution. Recovering the second from the first is the central problem of this literature.

Source: Dell, Jones & Olken (2014), *JEL*, Section 1

# How can we measure it?

WHAT WE WANT

$$y = f(C, X)$$

the response of $y$ to the climate $C$ — which varies across places, and so does everything in $X$

Source: Dell, Jones & Olken (2014), *JEL*, Sections 2.1–2.2

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COMPARES PLACES

**The cross-section**

different places with different climates,
at one point in time

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|---|---|
| HOLDS FIXED | nothing. $X$ has to enter as a control. |
| ASSUMES | no omitted term correlated with $C$; and that $C$ did not itself shape $X$. |
| IDENTIFIES | a long-run response, with adaptation already in it. |

Source: Dell, Jones & Olken (2014), *JEL*, Sections 2.1–2.2

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Source: Dell, Jones & Olken (2014), *JEL*, Sections 2.1–2.2

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THE TRADE-OFF The cross-section has the right variation but the wrong identification; the panel has the right identification but arguably the wrong variation. The next frames take each one in turn.

Source: Dell, Jones & Olken (2014), *JEL*, Sections 2.1–2.2

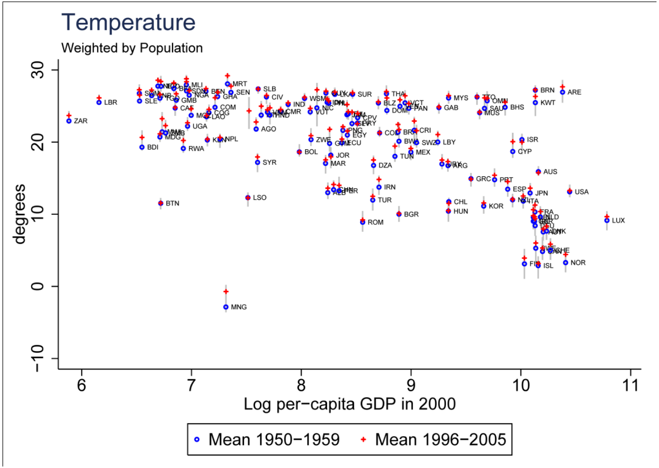
# The cross-sectional approach

$$y_i = \alpha + \beta C_i + \gamma X_i + \varepsilon_i$$

- The underlying correlation is large and well documented: across countries in 2000, a climate 1 °C warmer is associated with income per capita roughly 8.5% lower.
- What could go wrong?
  - **Omitted variables.** Countries with hot climates also differ in colonial history, disease environment and institutional quality. If those differences enter $\varepsilon_i$, then $\mathrm{Cov}(C_i, \varepsilon_i) \neq 0$ and $\hat{\beta}$ is inconsistent.
  - **Over-controlling.** Suppose instead that $X = X(C)$, so that climate itself shaped those institutions. Conditioning on $X$ then removes part of the effect we are trying to measure.

Source: Dell, Jones & Olken (2014), *JEL*, Section 2.1

# The cross-sectional correlation




- One point per country: population-weighted mean temperature against log GDP per capita in 2000.
- Circles are the 1950–1959 mean, crosses the 1996–2005 mean, the grey line the range of annual values in between.
- Mauritania at 28.4 °C, Mongolia at 1.77 °C; the slope is −0.085 log points per degree.
- The grey lines are short next to the spread across countries. That gap is what the panel design has to live on.

Source: Dell, Jones & Olken (2012), *AEJ: Macro*, Figure 1 (temperature panel)

# The panel approach

$$y_{it} = \beta C_{it} + \mu_i + \theta_{rt} + \varepsilon_{it}$$

- The country fixed effects $\mu_i$ absorb all time-invariant determinants of the outcome, including institutions, geography, colonial history and the average climate itself.
- The region-by-year fixed effects $\theta_{rt}$ absorb shocks common to a group of countries in a given year, such as global commodity cycles.
- $\beta$ is therefore identified from year-to-year deviations of a country from its own long-run average, measured relative to other countries in the same region and year.

**Identifying assumption**

Conditional on $\mu_i$ and $\theta_{rt}$, the weather realisation $C_{it}$ is uncorrelated with $\varepsilon_{it}$. This assumption is unusually plausible by the standards of applied macroeconomics: a country's policies and economic conditions do not determine its own annual weather deviations.

Source: Dell, Jones & Olken (2014), *JEL*, Section 2.2

# Sources of weather data

| Type | Examples |
|---|---|
| Ground stations | GHCN-Daily |
| Interpolated grids | CRU, UDEL ($0.5^\circ \times 0.5^\circ$), PRISM |
| Satellite | UAH, RSS ($2.5^\circ$), available from 1979 |
| Reanalysis | NCEP, ERA5-Land ($0.1^\circ$, hourly) |

- None of these is "the temperature". Stations observe, grids interpolate between stations, satellites measure the atmosphere rather than the surface, and reanalysis is a weather model fitted to whatever observations exist.
- Coverage is not stable over time either: the number of reporting GHCN stations falls sharply around 1990 with the dissolution of the USSR, which is a break in the data rather than in the climate.
- Papers routinely report results for more than one product, and sensitivity to that choice is a warning sign.

Source: Dell, Jones & Olken (2014), *JEL*, Section 2.3 and Figure 1

# How to model climate variables?

Between a gridded weather product and the regressor $C_{it}$ there are three decisions, none of which is innocuous.

1. **Which product.** Stations, interpolated grids, satellite or reanalysis. Products disagree with one another, and they disagree more about rainfall than about temperature.
2. **How to aggregate space.** A grid cell is not a country. The cells have to be averaged into the unit of analysis, and the weights used — area or population — are a substantive choice rather than a technical one.
3. **Which functional form, and over what period.** Whether temperature enters linearly, in bins, as an anomaly or as degree-days, and whether it is averaged before or after that transformation.

Source: Dell, Jones & Olken (2014), *JEL*, Sections 2.3–2.4

# Aggregation and measurement error

## Aggregating grid cells to a country

Area weights treat uninhabited territory and dense urban areas equally; population weights instead weight by exposure. For the United States in 2000 the two give

$$\bar{T}^{\text{area}} = 8.3\,^{\circ}\mathrm{C} \qquad \text{and} \qquad \bar{T}^{\text{pop}} = 13.1\,^{\circ}\mathrm{C}$$

a difference of 4.8 °C, larger than a century of observed warming.

## Precipitation is measured less reliably than temperature

Temperature is spatially smooth; rainfall is not. The correlation of deviations across data products is 0.92 for temperature but only 0.70 for precipitation. A null result on rainfall may therefore reflect attenuation from measurement error rather than the absence of an effect.

Source: Dell, Jones & Olken (2014), *JEL*, Section 2.3

# Functional form

1. **Levels**: $\beta\, T_{it}$. A single parameter, which imposes that the marginal effect of one degree is the same at 5 °C and at 35 °C.
2. **Bins**: $\sum_k \beta_k\, T_{kit}$, where $T_{kit}$ counts days spent in temperature range $k$. This imposes no shape on the response and is the specification discussed in Part III.
3. **Anomalies**: $(T_{it} - \bar{T}_i)/\sigma_i$. Comparable across places with very different climates, at the cost of a normalisation.
4. **Degree-days**: $\int \max(0, T - \underline{T})\, dt$, motivated by crop biology. Schlenker and Roberts (2009) estimate thresholds of 29 °C for corn, 30 °C for soybeans and 32 °C for cotton.

## Temporal aggregation

If temperature is averaged before being binned, the nonlinearity of interest is averaged away: a month with a mean of 24 °C may contain a week above 32 °C or none at all.

Source: Dell, Jones & Olken (2014), *JEL*, Section 2.4

# Part II

Temperature shocks and economic growth

# Dell, Jones & Olken (2012)

## Research question

Does a temperature shock reduce the *level* of output in the year in which it occurs, or the *rate of growth*, so that output never returns to its previous path? The two are difficult to tell apart in a single year and imply damages that differ by orders of magnitude over a century.

- **Setting.** All countries with population above one million, 1950–2003, annual; $n = 4{,}924$ country-years.
- **Data.** Temperature and precipitation from Matsuura and Willmott ($0.5^\circ$ grid, population-weighted to the country); GDP per capita from the World Development Indicators, with the Penn World Table as a robustness check.
- **Design.** A distributed lag in temperature with country fixed effects, region-by-year and poor-by-year fixed effects, and standard errors clustered by country and by region-year.
- **Headline result.** An increase of $1\,^\circ$C reduces growth in poor countries by about 1.4 percentage points, and the loss is not recovered in subsequent years. No effect is detected in rich countries.

Source: Dell, Jones & Olken (2012), *AEJ: Macro*

# Levels versus growth

### A level effect

A hot year reduces output in that year. In the following year, with normal weather, output returns to its previous trend. The growth path is unchanged.

### A growth effect

A hot year reduces the rate at which capital, productivity or institutional capacity accumulate. Output does not return to its previous path, which is permanently lower.

- Integrated assessment models almost always make this choice by assumption, writing damages as $D(T)\,A_t F(K, L)$: temperature scales the level of output, and growth is left untouched.
- The alternative is $\log A_t = \log A_{t-1} + g(T)$, in which a hot year lowers the whole subsequent path. Here the choice between the two is treated as a hypothesis to be tested rather than a modelling convention.
- Over a century the difference is not a technicality: it is the difference between damages of a few per cent of output and damages several times larger.

Source: Dell, Jones & Olken (2012), *AEJ: Macro*, Section I

# A model in which both channels are present

Let temperature affect both the level of productivity and its rate of change:

$$Y_{it} = e^{\beta T_{it}} A_{it} L_{it}, \qquad \frac{\Delta A_{it}}{A_{it}} = g_i + \gamma T_{it}$$

Taking logarithms and first differences gives growth in output per capita as

$$\boxed{g_{it} = g_i + (\beta + \gamma) T_{it} - \beta T_{it-1}}$$

- The level parameter $\beta$ enters contemporaneously with coefficient $+\beta$ and enters again in the following period with coefficient $-\beta$: its contribution to the sum of the coefficients is zero.
- The growth parameter $\gamma$ enters once and is not reversed.
- The sum of the distributed-lag coefficients therefore distinguishes the two channels, and this is what the empirical work exploits.

Source: Dell, Jones & Olken (2012), equations (1)–(3)

# The estimating equation

$$g_{it} = \mu_i + \theta_{rt} + \sum_{j=0}^{L} \rho_j \, T_{it-j} + \varepsilon_{it}$$

- $\mu_i$ denotes country fixed effects, so that identification comes from a country's own deviations from its mean.
- $\theta_{rt}$ denotes region-by-year and poor-by-year fixed effects. A poor country is therefore compared with other poor countries in the same year rather than with the OECD.
- Standard errors are clustered in two dimensions, by country and by region-year, following Cameron, Gelbach and Miller: weather is serially correlated within a country and spatially correlated across neighbours.
- First differencing is preferred to a levels specification with a lagged dependent variable because of Nickell bias; Bond, Leblebicioğlu and Schiantarelli (2010) show that with $T$ large the first-differenced estimator is preferable in growth regressions.
- "Poor" is defined as below-median PPP GDP per capita in the country's first year in the sample. The classification is fixed, so it cannot itself be an outcome of temperature.

Source: Dell, Jones & Olken (2012), equation (4)

# Hypotheses

**Without lags ($L = 0$): $H_0 : \rho_0 = 0$**

Temperature has no contemporaneous association with growth.

**With lags ($L > 0$): $H_0^1 : \rho_0 = 0$**

Temperature has no immediate effect once dynamics are allowed for.

**With lags ($L > 0$): $H_0^2 : \sum_{j=0}^{L} \rho_j = 0$**

This is the hypothesis of interest. Under a pure level effect the lagged terms offset the contemporaneous term and the sum is zero. Rejecting $H_0^2$ indicates that the shock is not recovered, which is the signature of a growth effect.

$H_0^2$ is a restriction on a linear combination of coefficients rather than on a single coefficient, which is why the covariance structure of the errors matters.

# Main results

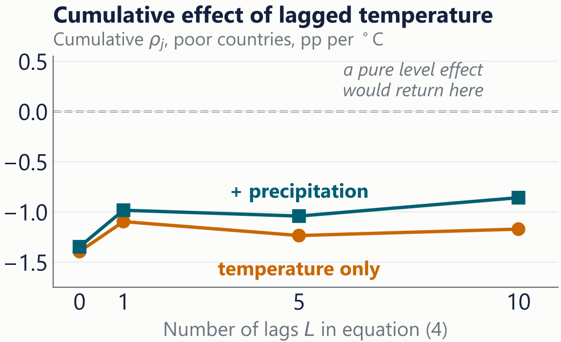
Dependent variable: growth in GDP per capita, 1950–2003, $n = 4{,}924$.

| Panel A. Effect of $+1\,^{\circ}$C on annual growth (pp) | |
|---|---|
| Temperature, all countries pooled | $-0.325$ (0.285) |
| Temperature, rich countries | 0.261 (0.312) |
| Temperature $\times$ Poor | $-1.655^{***}$ (0.485) |
| **Implied effect in poor countries** | $-\mathbf{1.394}^{***}$ (0.408) |
| with precipitation controls | $-1.347^{***}$ (0.408) |
| **Panel B. Alternative interactions** | |
| including Temperature $\times$ Hot | $-1.473^{***}$ |
| including Temperature $\times$ Agricultural | $-1.245^{***}$ |
| Precipitation, poor countries (per 100 mm) | 0.069 (0.058) |
| Precipitation, rich countries | $-0.083^{*}$ (0.050) |

One standard deviation of the temperature shock is $0.50\,^{\circ}$C, implying a cost of about 0.69 percentage points of growth. Panel B shows that interactions with "hot" or "agricultural" do not displace the interaction with income.

Source: Dell, Jones & Olken (2012), *AEJ: Macro*, Table 2

# Cumulative effects of lagged temperature




Dell, Jones & Olken (2012), Table 3. Rich countries: 0 throughout.

- Under a pure level effect, $\sum_j \rho_j$ should approach zero as lags are added (dashed line).
- The estimated sums do not. At ten lags they remain at $-1.17$ without precipitation controls and $-0.86$ with them.
- For rich countries the sums are indistinguishable from zero at every lag length.
- $H_0^2$ is therefore rejected for poor countries: the effect is on growth, not only on levels.

On precision: with ten lags and two-way clustering the estimates are imprecise, and the ten-lag sum with precipitation controls is no longer significant at the 5% level. The evidence is the pattern across lag lengths, not any single column.

Source: Dell, Jones & Olken (2012), *AEJ: Macro*, Table 3

# Robustness

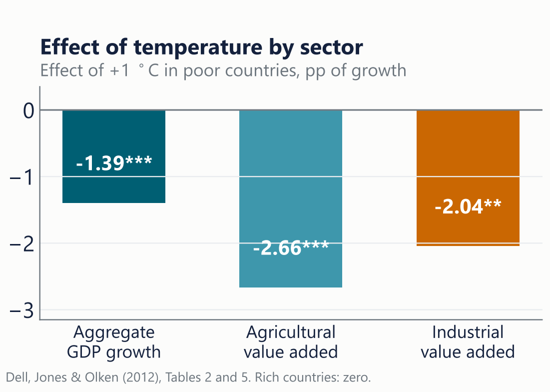
Immediate effect of $+1\,^\circ\mathrm{C}$ on growth in poor countries, percentage points.

| **Specification** | **Estimate** |
|---|---|
| Baseline, with precipitation controls | $-1.347^{***}$ |
| Including country-specific time trends | $-1.723^{***}$ |
| Balanced panel | $-1.377^{***}$ |
| Wider sample, including small countries | $-1.158^{***}$ |
| Penn World Table income in place of WDI | $-0.860^{***}$ |
| Area-weighted rather than population-weighted temperature | $-0.891^{**}$ |
| Sub-Saharan Africa only | $-1.881^{***}$ |
| Sub-Saharan Africa excluded | $-0.904^{*}$ |

The point estimate ranges between roughly $-0.9$ and $-1.9$ across specifications, while the sign and statistical significance are stable. The last two rows indicate that the result is not driven by Sub-Saharan Africa.

Source: Dell, Jones & Olken (2012), *AEJ: Macro*, Table 4

# Sectoral decomposition




- Agricultural value added responds most strongly, which is the expected result.
- Industrial value added falls by 2.04 percentage points per 1 °C, which is not readily explained by an agricultural mechanism.
- The estimated effect on investment is negative but not statistically significant.
- Precipitation has little measurable effect even in agriculture (+0.18 percentage points per 100 mm, not significant).
- The breadth of the response across sectors is part of what supports the growth interpretation.

Source: Dell, Jones & Olken (2012), *AEJ: Macro*, Table 5

# Summary of Part II

1. A model in which temperature affects both the level and the growth of productivity generates distinct predictions for the distributed-lag coefficients.
2. The specification is a distributed lag with country and region-by-year fixed effects and two-way clustered standard errors.
3. An increase of $1\,^{\circ}\mathrm{C}$ reduces growth in poor countries by about 1.4 percentage points. For rich countries no effect is detected.
4. The sum of the lag coefficients does not return to zero, which is consistent with an effect on growth rather than on levels.
5. The response appears in agriculture, in industry and in the probability of an irregular leadership transition.
6. Medium-run estimates are no smaller, but the seven-year calibration shows that the relationship cannot be extrapolated over a century.

# Part III

With or without U? Binning bias

# Jones, Moscona, Olken & von Dessauer (2026)

## Research question

Binned panel regressions consistently produce a U-shaped response of economic outcomes to temperature, with both extremes appearing damaging. Is that shape a property of the economy, or is part of it produced by the specification itself?

- **Setting.** United States counties, with four outcomes taken from four established literatures: population (1970–2010), mortality above age 65 (1970–2002), corn yields (1985–2006) and violent crime (1960–2009).
- **Data.** ERA5-Land reanalysis, $0.1^\circ$ and hourly from 1970, assigned to counties and counted into ten bins of $10\,^\circ$F.
- **Design.** The bias is derived analytically, confirmed in simulations in which the true effect of every bin is zero by construction, and then a correction is proposed and applied to the four literatures.
- **Headline result.** Of the four published patterns, one is reversed, one substantially weakened, one strengthened, and one unaffected.

Source: Jones, Moscona, Olken & von Dessauer (2026), NBER Working Paper 34671

# The binned specification

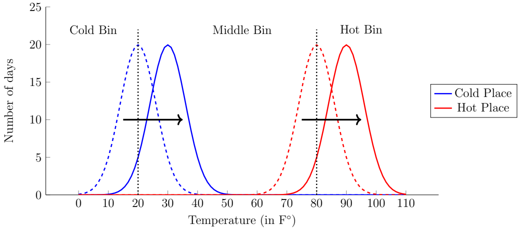
$$Y_{it} = \alpha_i + \delta_t + \sum_k \beta_k T_{kit} + u_{it}$$

- $T_{kit}$ counts the days that unit $i$ spends in temperature bin $k$ during year $t$; a common choice is ten bins of 10 °F, from below 10 °F to above 90 °F.
- The specification imposes no functional form, which is why it was described earlier as the appropriate way to estimate a nonlinear response.
- Estimates obtained this way are strikingly consistent across outcomes: a U-shape for mortality, crime and energy demand, and an inverted U for output and agricultural yields. Extreme temperatures appear damaging in both directions.
- These coefficients enter damage functions directly, and through them the social cost of carbon.

The argument of this paper is that part of the estimated U-shape is an artefact of warming itself.

Source: Jones, Moscona, Olken & von Dessauer (2026), equation (1)

# An illustration: Boston and Phoenix




Warming of the same size moves both distributions equally, but only the hot place has days to push into the top bin, and only the cold place has cold days left to lose.

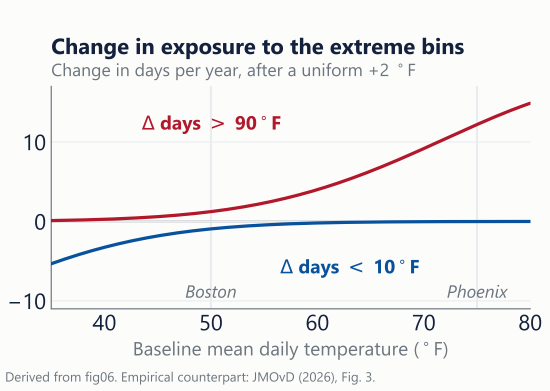
The same asymmetry in observed data: days per decade in the extreme bins, 1970s against 2010s.

| | **Days below 10 °F** | | **Days above 90 °F** | |
|---|---|---|---|---|
| | 1970s | 2010s | 1970s | 2010s |
| Boston | 20 | 10 | 0 | 0 |
| Phoenix | 0 | 0 | 1,000 | 1,062 |

The trend in an extreme bin is therefore a property of baseline climate, not of response to heat.

Source: Jones, Moscona, Olken & von Dessauer (2026), Figure 1 and Table 1; ERA5-Land

# Why exposure to the extreme bins trends unevenly




If daily temperature is distributed $\mathcal{N}(\mu_i, \sigma^2)$ and warming shifts the mean by $\delta$, then to a first approximation

$$\Delta\,\text{days}_{<10} \approx -365\,\phi\!\left(\frac{10-\mu_i}{\sigma}\right)\frac{\delta}{\sigma}$$

- Both changes are increasing in the baseline mean $\mu_i$: warmer places gain more hot days and lose fewer cold days.
- No economic behaviour is involved; this follows from the shape of the distribution.
- Across US counties both relationships are observed, positive and significant at $p \leq 0.001$, while the moderate bins are flat.

Source: Derived here; empirical counterpart in JMOvD (2026), Figure 3

# The source of the bias

Suppose the true model contains a place-specific trend that the researcher does not include:

$$Y_{it} = \alpha_i + \delta_t + \sum_k \beta_k T_{kit} + \underbrace{\lambda_i t}_{\text{omitted}} + \varepsilon_{it}$$

and let the bin counts themselves trend at place-specific rates:

$$C_{it} = \mu_{C,i} + \varphi_{C,i}\, t + v_{C,it}, \qquad H_{it} = \mu_{H,i} + \varphi_{H,i}\, t + v_{H,it}$$

The researcher estimates the model without $\lambda_i t$, so that the error term is $u_{it} = \lambda_i t + \varepsilon_{it}$.

## Result 1

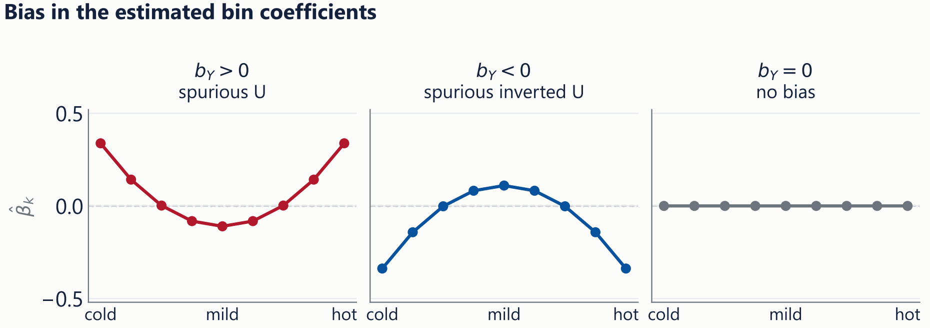
The bias in $\hat{\beta}_C$ and $\hat{\beta}_H$ is a function of the variances and covariances of $(\varphi_C, \varphi_H, \lambda)$ across places. This is omitted-variable bias in its standard form. What is specific to this setting is that the omitted variable is time itself, and its correlation with the regressor is guaranteed by the physical argument on the previous slide.

Source: Jones, Moscona, Olken & von Dessauer (2026), Section 2, equations (2)–(8)

# Result 2: the shape of the bias

Let each trend be linear in the location's baseline temperature $T_{0,i}$:

$$\varphi_{C,i} = a_C + b_C T_{0,i} + e_{C,i}, \quad \varphi_{H,i} = a_H + b_H T_{0,i} + e_{H,i}, \quad \lambda_i = a_Y + b_Y T_{0,i} + e_{Y,i}$$




Schematic of Result 2 in Jones, Moscona, Olken & von Dessauer (2026); the true effect of every bin is zero.

The sign of $b_Y$ asks whether the outcome trends differently in places that were already hot.

Source: Jones, Moscona, Olken & von Dessauer (2026), Result 2

# Four implications

1. The bias is increasing in $b_Y$. Any process that causes hot places to trend differently — migration to the Sun Belt, the diffusion of air conditioning, irrigation, divergent regional growth — will generate it.
2. The bias is decreasing in the noise in the trends. Idiosyncratic variation in local trends is protective, so systematic and precisely estimated trends are the more dangerous case.
3. Longer panels make the problem worse. As $T$ increases, the deterministic component of the bin counts grows relative to the idiosyncratic year-to-year variation, so a larger share of the identifying variation is trend. Adding years of data does not attenuate the bias.
4. Weak but precisely estimated trend relationships are the least favourable case, producing large and statistically significant coefficients with no underlying effect.

In simulations calibrated so that the true effect of every bin is zero, the top and bottom bins reject the null far more often than 5% of the time, and the estimated response resembles those reported in the published literature.

Source: Jones, Moscona, Olken & von Dessauer (2026), Sections 2 and 4

# Spurious evidence of adaptation

- The conventional test for adaptation compares hot and cold counties and asks whether hot counties are less affected by hot days. A smaller estimated effect in hot counties is read as evidence that agents have adapted.
- In the simulation, where by construction no agent adapts because no bin has any effect, exactly this pattern emerges.

| **Simulated bias in** | **Hot counties** | **Cold counties** |
|---|---|---|
| the cold bin (below 10 °F) | 28.58 | 0.72 |
| the hot bin (above 90 °F) | 1.60 | 6.41 |

The asymmetry is not driven by steeper trends in these subsamples but by lower noise: cold counties have very stable counts of cold days, so the systematic component dominates.

Source: Jones, Moscona, Olken & von Dessauer (2026), Figure 6 and Table A.2

# A correction based on counterfactual exposure

For each place and year, construct the expected number of days in each bin under a counterfactual distribution $F_{ct}$ containing only the systematic trend:

$$\bar{T}_{k,ct} \;=\; 365 \times \big(F_{ct}(\bar{x}) - F_{ct}(\underline{x})\big)$$

and include it in the regression alongside the realised counts:

$$Y_{ct} \;=\; \alpha_c + \delta_t + \sum_k \beta_k T_{kct} + \sum_k \gamma_k \bar{T}_{k,ct} + u_{ct}$$

- $\beta_k$ is then identified from the unexpected component of exposure, which is weather in the sense used at the start of the class, rather than from the trend.
- The logic is that of a control function (Wooldridge 2015), and it is the same idea as in Borusyak and Hull (2023) on non-random exposure to exogenous shocks: the shock is recentred on its expectation.

Source: Jones, Moscona, Olken & von Dessauer (2026), equations (20)–(21)

# Constructing the counterfactual distribution

1. For each county $c$ and calendar month $m$, regress daily mean temperature on a linear time trend: $\bar{x}_{cmt} = \mu_{cm} + \varphi_{cm} t + \varepsilon_{cmt}$.
2. Shrink the estimated trends towards the national mean using an empirical Bayes adjustment, so that noisily estimated county-months do not drive the counterfactual.
3. Remove the trend from every daily observation: $x_{cdmt}^{\text{detrended}} = x_{cdmt} - \hat{\varphi}_{cm}\, t$.
4. Pool the de-trended observations within a county-month. For January in a single county this gives approximately $31 \times 50 = 1{,}550$ draws, which serve as an estimate of the shape of the local distribution.
5. Project the distribution forward using $\hat{\varphi}_{cm}$ to obtain $F_{cmt}$, aggregate months to a county-year distribution $F_{ct}$, and compute the expected bin counts from equation (20).

The procedure uses only the temperature data, so it can be reproduced in any application; the authors distribute it as the Stata package `cftemp`.

Source: Jones, Moscona, Olken & von Dessauer (2026), Section 5.1, equations (22)–(23)

# Alternative strategies

| Strategy | Corrects the bias? | Assessment |
|---|---|---|
| State-by-year fixed effects | No | The top and bottom bins still reject falsely at 26% and 14%. The apparent improvement comes from added noise rather than reduced bias. |
| Five lags of $Y$ and of $X$ | No | False rejection rates exceed 90%. Lags absorb common dynamics but not place-specific ones. |
| Place-specific linear trends | Partly | Adequate when the outcome trend is genuinely linear; fails under quadratic or cubic trends, and requires more than 3,000 additional parameters. |
| County-by-five-year-period fixed effects | Partly | Removes the bias in simulation, including with nonlinear trends, but requires annual data and is therefore unavailable for decadal outcomes or long differences. |
| Counterfactual bin counts, eq. (21) | Yes | Estimates are centred at zero when there is no effect and recover the true effect when there is one. The correction operates on the right-hand side, which is why it generalises across outcomes. |

Source: Jones, Moscona, Olken & von Dessauer (2026), Sections 5.1–5.2

# Applications

| Outcome | Uncorrected estimate | After correction |
|---|---|---|
| **Population** (Census, 1970–2010) | U-shape, implying movement towards both temperature extremes | The response flattens and the top bin becomes significantly negative, which is consistent with migration away from extreme heat. |
| **Mortality** (age 65+, 1970–2002) | The standard U: both cold and heat raise mortality | Cold-bin coefficients fall by more than half and the effect above 90 °F falls to zero. Standard errors also decline. |
| **Corn yields** (NASS, 1985–2006) | Pronounced inverted U, with extreme heat reducing yields | Essentially unchanged. Yield trends are not correlated with baseline temperature, so $b_Y \approx 0$, as Result 2 predicts. |
| **Violent crime** (FBI, 1960–2009) | Heat raises crime; cold reduces it weakly | The heat effect weakens while the cold effect strengthens substantially. |

The correction does not overturn the literature uniformly: it alters some estimates substantially, leaves others unchanged, and can be applied at low cost in any given application.

Source: Jones, Moscona, Olken & von Dessauer (2026), Section 6 and Figure 12

# Summary of Part III

1. Uniform warming does not shift every location's temperature bins in the same way. How much exposure to extreme heat a place gains depends on its baseline climate.
2. The regressor in a binned panel therefore carries a place-specific trend that is systematically related to baseline climate.
3. If the outcome also trends with baseline climate, so that $b_Y \neq 0$, the regression attributes that trend to the extreme bins and produces a U-shape that is not present in the data-generating process.
4. Longer panels aggravate the problem, and the resulting pattern can be mistaken for evidence of adaptation.
5. The proposed correction controls for expected exposure, constructed from the temperature data alone.
6. Applied to four literatures, one result is reversed, one substantially weakened, one strengthened, and one unaffected.

# Concluding remarks

1. Every estimate in this literature combines a research design with an identifying assumption. The cross-section assumes no confounders; the panel assumes that the short-run response is informative about the long run; the binned panel assumes that exposure does not trend.
2. The panel design is a genuine methodological advance, since weather is as close to random assignment as macroeconomic data permits. It nonetheless does not identify the effect of a change in climate.
3. Temperature affects the growth rate, and not only the level, of output in poor countries: approximately 1.4 percentage points per 1 °C, with effects in agriculture, in industry and in political stability.
4. Established empirical regularities may be artefacts of the specification. When the environment itself is trending, the properties of the estimator have to be re-derived.
5. In the next class we construct one of these panels: gridded weather data, a shapefile, aggregation to administrative units, estimation, and mapping, in R.

# References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- Jones, B. F., J. Moscona, B. A. Olken & E. von Dessauer (2026). "With or Without U? Binning Bias and the Causal Effects of Temperature Extremes." NBER Working Paper 34671.
- Also cited: Schlenker & Roberts (2009); Deschênes & Greenstone (2007, 2011); Fisher, Hanemann, Roberts & Schlenker (2012); Barreca et al. (2016); Burke, Hsiang & Miguel (2015); Hsiang, Meng & Cane (2011); Hornbeck (2012); Ranson (2014); Borusyak & Hull (2023); Pindyck (2013).
- Weather data: GHCN-Daily; Matsuura & Willmott (UDEL); CRU; PRISM; ERA5-Land (Muñoz-Sabater et al. 2021), which is used in the next class.
- Code: `cftemp`; `github.com/bolken44/counterfactual_temperatures`.


## Questions?